Name - Gurveer Singh Mangat
UID - 23BCS11074
Section - 23Krg 2B

Sr. No. Date:

Q ⇒ Given three integers n, a and b return the nth magical no.. Ans may be large may be very large return 10^9+7

Magical no.: if no either divisible by a to b

Ex - n = 1, a = 2, b = 3
out - b

⇒ Approach

i) Find LCM of a and b
lcm = (a*b) / gcd (a, b)

ii) Set the range for searching using binary search.
low = min (a, b)
high = n* min (a, b)

iii) Now next find the mid ⇒ low + (right - low)/2

iv) for each Mid count Magical no. (while low<high)
Magic Count = mid/a + mid/b − mid/lcm

If Magic Count < n
low = mid + 1
else
high = mid − 1

v) return low % (10^9+7)

Sr. No

TC – $O(\log(n * \min(a, b)))$

* Brute force

i) Make an ~~array~~ array ~~ordered~~ set of elements and unordered m

ii) Push elements of 2 and 3 till the size is equal to n-1
→ Mark each element in the map and if it occurs again don't push it in the array

iii). This leads to a common multiples. So we don't push them

iv) return the last value of array.